



Courtesy by DRDL

PLAYING WITH FIRE

The first flight of Agni was aborted in 1989 on the launchpad to make some corrections which took nearly a month. Kalam held a gate meeting and told the staff that the second test would be within a month. It was.

ALL SMILES: Kalam after the Agni launch in 1989

DEFENCE RESEARCH AND DEVELOPMENT LABORATORIES, HYDERABAD

KALAM CONTROL

The scientists and engineers of the Defence Research and Development Laboratories (DRDL) breathed a sigh of relief when A.P.J. Abdul Kalam became the director of the seemingly innocuous organisation in 1982. He brought a new commitment to indigenous research and spawned the Integrated Guided Missile Development Programme (IGMDP)—the most successful Indian military research task to-date that sought to develop a clutch of strategic and guided missiles.

The DRDL's list of accomplishments were to get longer as time passed. It joined hands with the Nizam Institute of Medical Sciences to develop the Kalam-Raju stent in 1998, the first made-in-India stent to be used by cardiac surgeons, and later to mould composite materials to make the lighter calipers worn by thousands of the physically challenged today. But the turning point came when it made a foray into missile building that eventually transformed from exercises in technology-gathering, reverse-engineering, and design competence, to fruition with the IGMDP that is now helping build a series of operational missile systems. Kalam involved several organisations to carry out various tasks directly related to the IGMDP. Through con-

current engineering, the period required to execute the most challenging project ever undertaken in the country was reduced to the minimum. But that wasn't all. Unfazed by the constraints imposed by the Missile Technology Control Regime in buying a CRAY-XMP super-computer which had a 55 mega flops capability, Kalam and the DRDL steered the initiative to establish an exclusive facility to develop a fast processing computer within three years.

DRDL is a pioneer in designing digital weaponry. It has developed the long-range surface-to-air missile with the second biggest weapons supplier, Israel. It has expanded its infrastructural facilities to take up new programmes like the one to develop hypersonic cruise vehicles which would fly at speeds of 6,000-7,000 kmph. The US, the UK, France, Russia, Australia and China are pursuing hypersonic technology. Even as it joins such elite clubs to push the frontiers of technology, DRDL's long-range power projection programme is the key to maintaining strategic stability in the Asia-Pacific region.

by Amarnath K. Menon

"We are a systems laboratory for tactical missile programmes. A pool of professionals, we've proved we can do things on our own."

P. VENUGOPALAN, DIRECTOR, DRDL